Exam 2 In-Class Review

Module 6 - Multiple Dosings · Review set · 7 Oct 2026

Name _____ Date _____

Question 1 Using the patient's ideal body weight, estimate the creatinine clearance of a 45-year-old female who weighs 154 lb, serum creatinine of 0.78 mg/dL, and stands 5'8" tall.

Question 2 A drug was administered as a single IV bolus dose of 10 mg/kg to a 75-kg male volunteer. The drug has an elimination half-life of approximately 5 hours and volume of distribution of 2.3 L/kg. Urine samples were collected for 48 hours and upon analysis, 630 mg was recovered. What is the renal clearance in mL/min and probable mechanism of renal elimination?

Question 2 A drug was administered as a single IV bolus dose of 10 mg/kg to a 75-kg male volunteer. The drug has an elimination half-life of approximately 5 hours and volume of distribution of 2.3 L/kg. Urine samples were collected for 48 hours and upon analysis, 630 mg was recovered. What is the probable mechanism of renal elimination?

A. Filtration with tubular reabsorption
B. Filtration with active secretion
C. Filtration only

3a An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. a. What is the expected maximum steady-state concentration of this antibiotic in this patient?

3b An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. b. What dose would yield an average steady-state concentration of approximately 80 mcg/mL?

3c An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. c. If the AUC following a 500 mg oral dose of the antibiotic was 202 (mg/L)hr, what is the oral bioavailability of this antibiotic?

3d An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. d. What oral dose would provide the same extent of absorption as the 12 mg/kg IV dose?

3e An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. e. What C_{max} would you expect following the administration of the first oral dose in the amount you determined in part d if the absorption half-life is 75 minutes?

3f An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. f. What dosing interval would be appropriate to achieve an average steady-state concentration of 80 mcg/mL using the oral dose determined in part d?

3g An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. g. What is t_{max} at steady state if the dose determined in part d were administered per the dosing interval determined in part f?

3h An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. h. What is C_{max} if the dose determined in part d were administered per the dosing interval determined in part f?

Question 4 Four hundred milligrams of an antibiotic was infused intravenously over a period of 2 hours. Eight hours after the start of the first infusion, a second 400-mg dose was infused again over a period of 2 hours. The half-life of the antibiotic is approximately 3.75 hours and the apparent volume of distribution is approximately 22 L. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

Question 5 What is creatinine clearance used to estimate?

A. renal function
B. metabolic function
C. tubular reabsorption
D. apparent volume of distribution

Question 6 Which of the following would you expect with a 2-fold increase in oral dose? (Select all that apply)

A. Decreased AUC
B. Increased AUC
C. Decreased t_{max}
D. Decreased C_{max}
E. Increased t_{max}
F. Increased C_{max}

Question 7 A new antibiotic is excreted by the kidney and has a renal clearance that is approximately 300 mL/min. What is the probable mechanism of renal excretion of this drug?

A. Filtration with tubular reabsorption
B. Filtration with active secretion
C. Filtration only

Question 8 Which term describes the rate and extent to which the active ingredient or active moiety is absorbed from a drug product and becomes available at the site of action?

A. pharmacodynamics
B. bioavailability
C. product performance
D. bioequivalence

Question 9 Which of the following would be expected following a decrease in the dosing interval with no change to the dose?

A. increased steady-state concentrations
B. increased patient compliance
C. increased fluctuations between peak to trough concentrations
D. decreased MTC

Question 10

The principle of superposition assumes that early doses of drug have no effect on the pharmacokinetics of subsequent doses.

A. True
B. False

Question 11

Which term below describes a comparison of the bioavailability of the active drug in the systemic circulation following extravascular administration with the bioavailability of the same drug following intravenous administration?

A. drug product performance
B. bioequivalence
C. relative bioavailability
D. absolute bioavailability

Question 12

Which of the following is true regarding the absorption phase following oral administration?

A. drug at the absorption site has become depleted
B. the rate of drug absorption is faster than the rate of drug elimination
C. the rate of drug elimination is faster than the rate of drug absorption
D. the rate of drug absorption equals the rate of drug elimination

Question 13

When does the minimum concentration of drug in the plasma occur with multiple IV bolus dosings?

A. at the end of the dosing interval
B. in 3 to 5 half-lives
C. when the rate of absorption exceeds the rate of elimination
D. when the rate of absorption equals the rate of elimination

Question 14

If there is an decrease in the total body clearance of an agent due to renal dysfunction, what is the expected change to the elimination half-life of that agent?

A. the elimination half-life will increase
B. the elimination half-life will decrease
C. the elimination half-life will not change

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: 91.9 mL/min

WHAT THE STEM GIVES

age	45-year-old	140 − 45 = 95 in the numerator
sex	female	IBW constant 45.5 and the 0.85 factor
weight	154 lb	*not needed: the stem asks for ideal body weight*
S_{Cr}	0.78 mg/dL	72 × 0.78 = 56.16 in the denominator
height	5'8"	8 inches over 5 ft, so IBW = 63.9 kg

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{IBW (female)} = 45.5 + 2.3 \times (\text{inches over 5 ft}) = 45.5 + 2.3 \times 8 = 45.5 + 18.4 = 63.9 \text{ kg}$$

IBW, ideal body weight, is what the stem asks to use. 5'8" is 8 inches over 5 feet, and 45.5 is the female constant (50 for a male). The 154 lb actual weight is never used.

SETUP

$$\text{CrCl} = \frac{(140 - \text{age})(\text{IBW})}{72 \times S_{Cr}} \times 0.85 \text{ (female)}$$

Cockcroft-Gault, the line she says is not on the sheet and has to be known. Age in years, IBW in kg, S_{Cr} (serum creatinine) in mg/dL, and the 0.85 because the patient is female.

ALGEBRA

$$\frac{(140 - 45)(63.9)}{72 \times 0.78} = \frac{6070.5}{56.16} = 108.09$$

Top: 95 years × 63.9 kg = 6070.5. Bottom: 72 × 0.78 mg/dL = 56.16. The serum creatinine is used as given, 0.78; her slides state no rule that raises a value below 1.

ALGEBRA

$$\text{CrCl} = 108.09 \times 0.85 = 91.88 \text{ mL/min}$$

The female correction comes last. The result is reported in mL/min, the unit of the 120 to 130 mL/min normal range, although the units of the inputs do not cancel to it.

ROUND

$$\text{CrCl} = 91.9 \text{ mL/min}$$

91.88 to one decimal place. It is below the 120 to 130 mL/min she calls normal, so this patient has some loss of renal function.

> **DOES THE NUMBER MAKE SENSE?**
> With S_{Cr} below 1, the 72 × S_{Cr} denominator is below 72, so the male-form value exceeds IBW × 95 ÷ 72; 0.85 of 108.09 is 91.88.

Question 2 Worked by the drill

Answer: 334.7 mL/min

WHAT THE STEM GIVES

D_0	10 mg/kg	× 75 kg = 750 mg, the denominator of f_e
weight	75-kg	turns both per-kg values into mg and L
t½	approximately 5 hours	$k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	2.3 L/kg	× 75 kg = 172.5 L, for $Cl_T = kV_D$
collection	48 hours	9.6 half-lives, so the urine collection is complete
D_u	630 mg	numerator of $f_e = \frac{630}{750} = 0.84$

THE WORKING, ONE LINE AT A TIME

SETUP

$D_0 = 10\ mg/kg \times 75\ kg = 750\ mg$ (the dose); $V_D = 2.3\ L/kg \times 75\ kg = 172.5\ L$ (the volume)

Both are per kilogram, so each is multiplied by the 75-kg weight first. The kilograms cancel and leave mg and L.

UNIT

$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$ (elimination rate constant); $Cl_T = k \times V_D = 0.1386 \times 172.5 = 23.91\ L/hr$ (total body clearance)

Clearance is not given, so it is built from the half-life and the volume: $Cl = kV_D$, the line she expects without the sheet. $hr^{-1} \times L$ leaves L/hr.

ALGEBRA

$f_e = \frac{D_u}{D_0} = \frac{630\ mg}{750\ mg} = 0.84$ (fraction excreted unchanged)

D_u is the drug recovered unchanged in urine. 48 hours is 9.6 half-lives of 5 hours, so the collection is complete and 630 mg is all that will be excreted unchanged. F = 1 for an IV bolus.

ALGEBRA

$Cl_R = f_e \times Cl_T = 0.84 \times 23.91\ L/hr = 20.08\ L/hr$

Renal clearance is the excreted fraction of the total clearance: 84% of the drug leaves unchanged in urine, so 84% of the plasma cleared each hour is cleared by the kidney.

UNIT

$20.08\ L/hr \times \frac{1000\ mL}{1\ L} \times \frac{1\ hr}{60\ min} = 334.7\ mL/min$

The stem asks for mL/min, and the comparison with the 120 mL/min filtration rate needs mL/min: multiply by 1000 for mL, divide by 60 for minutes.

ROUND

Cl_R = 334.7 mL/min, above the GFR of about 120 mL/min: filtration with active secretion

Filtration alone clears at most about 120 mL/min, so 334.7 mL/min needs a second process carrying drug into the tubule. The second half of her question is this comparison.

DOES THE NUMBER MAKE SENSE?

Cl_R cannot exceed Cl_T, 23.91 L/hr; 0.84 of it is 20.08 L/hr, which is 334.7 mL/min.

Question 2

Worked by the drill

Answer: B. Filtration with active secretion

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = f_e \times Cl_T = 0.84 \times 23.91$ L/hr = 20.08 L/hr = 334.7 mL/min

$f_e = \frac{630}{750} = 0.84$ and $Cl_T = kV_D = 0.1386 \times 172.5 = 23.91$ L/hr, as in the calculation for this stem. The mechanism is read in mL/min.

ALGEBRA

334.7 mL/min against GFR ≈ 120 mL/min: greater

Above the filtration rate means a second process adds drug to the urine: active secretion. Filtration alone cannot clear more plasma than the glomeruli filter, about 120 mL/min.

The renal clearance works out to 334.7 mL/min, well above the glomerular filtration rate of about 120 mL/min. Filtration cannot clear more plasma than it filters, so a transporter must be carrying extra drug from the blood into the tubule.

3a

Worked by the drill

Answer: 109.1 mg/L

WHAT THE STEM GIVES

D_0	12 mg/kg	× 75 kg = 900 mg
τ	every 8 hours	kτ in the accumulation factor
weight	75-kg	turns mg/kg and % of body weight into mg and L
Cl_T	approximately 1.86 L/hr	$k = \frac{1.86}{11.25} = 0.16533$ hr^{-1}
V_D	15% of body weight	0.15 L/kg × 75 kg = 11.25 L

THE WORKING, ONE LINE AT A TIME

SETUP

D_0 = 12 mg/kg × 75 kg = 900 mg (each dose); V_D = 0.15 L/kg × 75 kg = 11.25 L (volume)

15% of body weight is read as 0.15 L/kg, 1 L taken as the weight of 1 kg. Both the dose and the volume are per kilogram, so each is multiplied by 75 kg and the kilograms cancel.

UNIT

$$k = \frac{Cl_T}{V_D} = \frac{1.86\ \text{L/hr}}{11.25\ \text{L}} = 0.16533\ \text{hr}^{-1}\ \text{(elimination rate constant)}$$

She gives clearance instead of a half-life here, so k comes from $Cl = kV_D$ rearranged. L cancels and leaves hr^{-1}.

ALGEBRA

$$C_0 = \frac{D_0}{V_D} = \frac{900\ \text{mg}}{11.25\ \text{L}} = 80\ \text{mg/L}\ \text{(what one dose adds)}$$

The first-dose peak of an IV bolus: the whole dose is in the body at time zero, spread through V_D. It is also what each later dose adds on top of what remains.

ALGEBRA

$k\tau = 0.16533 \times 8 = 1.3226$; $e^{-1.3226} = 0.2664$ (fraction left after one interval)

τ = 8 hr, the dosing interval. $k\tau$ has no units, because $hr^{-1} \times hr$ cancels, and $e^{-k\tau}$ is the fraction of each dose still in the body when the next one is given.

ALGEBRA

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}} = \frac{80}{1 - 0.2664} = \frac{80}{0.7336} = 109.05\ \text{mg/L}$$

The accumulation factor $\frac{1}{0.7336} = 1.36$ raises the first-dose peak to the steady-state peak, because each new dose lands on drug left from the earlier doses: the trough is $109.05 \times 0.2664 = 29.05$ mg/L.

ROUND

$$C_{max}^{\infty} = 109.1\ \text{mg/L}$$

109.05 to one decimal place; carrying every digit of k gives 109.055, which rounds to 109.1. The exam wants the answer with its units, here mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> The steady-state peak must exceed the first-dose peak of 80 mg/L and stay below twice it, because $e^{-k\tau} = 0.2664$ is under 0.5: 109.1 mg/L.

3b

Worked by the drill

Answer: 1190.4 mg

WHAT THE STEM GIVES

τ	every 8 hours	τ in $C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}$
Cl_T	approximately 1.86 L/hr	Cl_T in the denominator
C_{avg}^{∞}	approximately 80 mcg/mL	80 mg/L, the target
V_D	15% of body weight	*not needed: the average uses clearance only*
D_0	12 mg/kg	*not needed: this part finds a new dose*

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}$, solved for the dose: $D_0 = \frac{C_{avg}^{\infty} \times Cl_T \times \tau}{F}$

The average needs only clearance and the interval, not V_D or k, because over one interval at steady state the drug given, $F \times D_0$, equals the drug cleared, $C_{avg}^{\infty} \times Cl_T \times \tau$. F = 1 for an IV bolus.

UNIT

80 mcg/mL = 80 mg/L (target average)

1 mcg/mL equals 1 mg/L, because multiplying top and bottom by 1000 turns mcg into mg and mL into L. Her targets arrive in mcg/mL and her clearances in L/hr, so this conversion comes first.

ALGEBRA

$D_0 = 80\ mg/L \times 1.86\ L/hr \times 8\ hr = 148.8\ mg/hr \times 8\ hr = 1190.4\ mg$

mg/L × L/hr leaves mg/hr, the drug cleared each hour at the target; × 8 hr leaves mg, the drug to replace each interval. The same interval, every 8 hours, is kept.

ROUND

D_0 = 1190.4 mg every 8 hours (15.9 mg/kg for 75 kg)

In mg because the stem asks for a dose; $\frac{1190.4}{75}$ = 15.87 mg/kg if the answer is wanted per kilogram.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance removes 80 mg/L × 1.86 L/hr = 148.8 mg each hour at the target, so each 8-hour dose must replace 1190.4 mg.

3c

Worked by the drill

Answer: 0.75 (as a decimal fraction)

WHAT THE STEM GIVES

Cl_T	approximately 1.86 L/hr	gives $AUC_{IV} = \frac{500}{1.86} = 268.8$
D_{po}	500 mg oral dose	the dose in $F = \frac{Cl \times AUC}{D}$
AUC_{po}	202 (mg/L)hr	the oral area
D_0	12 mg/kg	*not needed for F*
V_D	15% of body weight	*not needed for F*

THE WORKING, ONE LINE AT A TIME

SETUP

AUC_{IV} for the same 500 mg $= \frac{D_0}{Cl_T} = \frac{500\ mg}{1.86\ L/hr} = 268.8$ (mg/L)hr

No IV area is given, so it is built from the clearance: $Cl = \frac{D_0}{AUC}$ rearranged. mg divided by L/hr leaves (mg/L)hr.

ALGEBRA

$$F_{abs} = \frac{AUC_{po}}{AUC_{IV}} \times \frac{D_{IV}}{D_{po}} = \frac{202}{268.8} \times \frac{500}{500} = 0.7515$$

Equal doses make the dose ratio 1, so F is the ratio of the areas. The same number comes from $F = \frac{Cl_T \times AUC_{po}}{D_{po}} = \frac{1.86 \times 202}{500} = 0.7514$.

ROUND

$F = 0.75$ (75%)

To the nearest hundredth, written 0.75 or 75%, never .75 (her rule: no leading decimals). F cannot exceed 1, and 0.75 means three quarters of the oral dose reached the circulation.

DOES THE NUMBER MAKE SENSE?

The oral area, 202, is below the IV area for the same dose, 268.8, so F is below 1: about three quarters.

3d

Worked by the drill

Answer: 1200 mg

WHAT THE STEM GIVES

D_{IV}	12 mg/kg	× 75 kg = 900 mg, the numerator
weight	75-kg	turns 12 mg/kg into 900 mg
F	part c	0.75, the denominator
τ	every 8 hours	*not needed: extent per dose only*

THE WORKING, ONE LINE AT A TIME

SETUP

$D_{po} = \frac{D_{IV}}{F}$ (same extent means the same amount reaching the blood: $F \times D_{po} = D_{IV}$)

Extent of absorption is the AUC. The oral dose has to put as much drug into the circulation as the IV dose, and only the fraction F of an oral dose gets there.

ALGEBRA

$D_{IV} = 12 \text{ mg/kg} \times 75 \text{ kg} = 900 \text{ mg}$; $D_{po} = \frac{900 \text{ mg}}{0.75} = 1200 \text{ mg}$

D_{IV} is the 12 mg/kg dose for 75 kg. F = 0.75 from part c; with F unrounded, 0.7514, the dose is 1197.7 mg. Either way the oral dose exceeds the IV dose.

ROUND

$D_{po} = 1200$ mg

Her solved sheets round an equivalent oral dose to a strength that exists; 1197.7 mg becomes 1200 mg. The oral dose is always larger than the IV dose because F is below 1.

DOES THE NUMBER MAKE SENSE?

F is below 1, so the oral dose must exceed the 900-mg IV dose: $\frac{900}{0.75} = 1200$ mg.

3e Worked by the drill

Answer: 47.8 mg/L

WHAT THE STEM GIVES

$t_{½a}$	75 minutes	1.25 hr, so $k_a = 0.5544\ hr^{-1}$
D_0	part d	1200 mg; with F = 0.75, 900 mg absorbed
Cl_T	approximately 1.86 L/hr	with V_D gives $k = 0.1653\ hr^{-1}$
V_D	15% of body weight	11.25 L in the front factor
τ	every 8 hours	*not needed: first dose only*

THE WORKING, ONE LINE AT A TIME

UNIT

$$k_a = \frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1}$$ (absorption rate constant); $k = 0.1653\ hr^{-1}$ (from part a)

75 minutes is 1.25 hours; both rate constants must be per hour before they are combined. $k = \frac{1.86}{11.25} = 0.1653\ hr^{-1}$ comes from the clearance and volume in the stem.

ALGEBRA

$$t_{max} = \frac{\ln(k_a \div k)}{k_a - k} = \frac{\ln(3.354)}{0.3891} = \frac{1.210}{0.3891} = 3.110\ hr$$ (time of the peak)

$k_a \div k = 0.5544 \div 0.1653 = 3.354$ and $k_a - k = 0.3891\ hr^{-1}$. C_{max} is the concentration at t_{max}, so t_{max} comes first.

ALGEBRA

$$\frac{F\ k_a\ D_0}{V_D(k_a - k)} = \frac{0.75 \times 0.5544 \times 1200}{11.25 \times 0.3891} = \frac{498.96}{4.3774} = 113.99\ mg/L$$

$F \times D_0 = 0.75 \times 1200 = 900$ mg reaches the blood. This front factor is not C_0, a time-zero concentration: at t = 0 the bracket is 1 − 1 = 0, so the oral curve starts at zero.

ALGEBRA

$$e^{-0.1653 \times 3.110} - e^{-0.5544 \times 3.110} = e^{-0.5141} - e^{-1.7242} = 0.5980 - 0.1783 = 0.4197$$

The bracket is evaluated at $t_{max} = 3.110$ hr: e^{-kt} carries elimination and $e^{-k_a t}$ carries absorption. Both exponents are pure numbers, because $hr^{-1} \times hr$ cancels.

ALGEBRA

$$C_{max} = 113.99 \times 0.4197 = 47.84\ mg/L$$

Front factor times bracket: 113.99 mg/L × 0.4197. The bracket has no units, so the answer keeps mg/L from the front factor.

ROUND

$$C_{max} = 47.8\ mg/L$$

With F unrounded, 0.7514 × 1200 = 901.7 mg absorbed, and the peak is 47.93 mg/L; the tolerance accepts both. This is the first dose, so no accumulation factor enters.

DOES THE NUMBER MAKE SENSE?

The bracket at t_{max} is below 1, so the peak lies below the front factor of 113.99 mg/L: 47.8 mg/L.

3f Worked by the drill

Answer: 6 hr

WHAT THE STEM GIVES

C_{avg}^{∞}	80 mcg/mL	80 mg/L in the denominator
D_0	part d	1200 mg; F × D_0 = 900 mg
Cl_T	approximately 1.86 L/hr	Cl_T in the denominator
V_D	15% of body weight	*not needed: the average uses clearance only*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}\text{, solved for the interval: } \tau = \frac{FD_0}{Cl_T \times C_{avg}^{\infty}}$$

The same average line as part b, with the dose fixed this time and the interval unknown. Multiplying both sides by τ and dividing by C_{avg}^{∞} isolates τ.

ALGEBRA

$$\tau = \frac{0.75 \times 1200\ \text{mg}}{1.86\ \text{L/hr} \times 80\ \text{mg/L}} = \frac{900}{148.8} = 6.05\ \text{hr}$$

80 mcg/mL = 80 mg/L. The top, 900 mg, is the drug absorbed per dose; the bottom, 148.8 mg/hr, is the drug cleared per hour at the target. mg divided by mg/hr leaves hr.

ROUND

τ = 6 hours

A dosing interval is given as a practical clock interval; her Practice 2 answer moved a regimen "from 8 to 6 hours". 6.05 hr becomes every 6 hours.

DOES THE NUMBER MAKE SENSE?

At the target, clearance removes 148.8 mg each hour, so 900 mg absorbed per dose lasts about 6 hours; a positive interval near 6 hours.

3g Worked by the drill

Answer: 2 hr

WHAT THE STEM GIVES

τ	part f	6 hr in both exponents
t½a	75 minutes	k_a = 0.5544 hr^{-1}
Cl_T	approximately 1.86 L/hr	with V_D gives k = 0.1653 hr^{-1}
D_0	part d	*not needed: t_{max} carries no dose*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max}^{\infty} = \frac{1}{k_a - k} \ln\left[\frac{k_a(1 - e^{-k\tau})}{k(1 - e^{-k_a \tau})}\right]$$

The steady-state time to peak, printed at the bottom of the right-hand column of her sheet. It needs k_a, k and τ; no dose, F or V_D.

ALGEBRA

$$k\tau = 0.1653 \times 6 = 0.9918,\ e^{-0.9918} = 0.3709;\ k_a\,\tau = 0.5544 \times 6 = 3.3264,\ e^{-3.3264} = 0.0359$$

τ = 6 hr from part f; k = 0.1653 hr^{-1} from part a and k_a = 0.5544 hr^{-1} from part e. Each exponent is a pure number, because hr^{-1} × hr cancels.

ALGEBRA

$$\frac{0.5544 \times 0.6291}{0.1653 \times 0.9641} = \frac{0.34877}{0.15937} = 2.1885;\ \ln 2.1885 = 0.7832$$

1 − 0.3709 = 0.6291 and 1 − 0.0359 = 0.9641. The ratio inside the logarithm has no units, because k_a and k are both in hr^{-1} and cancel top and bottom.

ALGEBRA

$$t_{max}^{\infty} = \frac{0.7832}{0.3891} = 2.013 \text{ hr}$$

$k_a - k = 0.5544 - 0.1653 = 0.3891\ hr^{-1}$, and a pure number divided by hr^{-1} leaves hours. The steady-state peak comes earlier than the first-dose peak, 3.110 hr.

ROUND

$$t_{max}^{\infty} = 2.0 \text{ hr}$$

With τ = 6.05 hr unrounded it is 2.02 hr; both round to 2.0 hr. No dose, F or V_D entered, which is why the dose and volume in the stem are not needed here.

DOES THE NUMBER MAKE SENSE?

The steady-state peak comes after the dose and before the end of the 6-hour interval, and earlier than a single-dose peak: 2.0 hr.

3h Worked by the drill

Answer: 91.2 mg/L

WHAT THE STEM GIVES

D_0	part d	1200 mg, F × D_0 = 900 mg
τ	part f	6 hr in the accumulation factor
V_D	15% of body weight	11.25 L
t½a	75 minutes	enters through t_{max}^{∞} from part g

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{FD_0}{V_D}\left(\frac{1}{1 - e^{-k\tau}}\right) e^{-k\, t_{max}^{\infty}}$$

Repeated oral doses at steady state: the amount absorbed per dose over V_D, times the accumulation factor, times the decline to the steady-state peak time.

ALGEBRA

$$\frac{FD_0}{V_D} = \frac{0.75 \times 1200}{11.25} = \frac{900}{11.25} = 80 \text{ mg/L}$$

F = 0.75 from part c and D_0 = 1200 mg from part d, so 900 mg reaches the blood with each dose; divided by V_D = 11.25 L it is the concentration one absorbed dose would give.

ALGEBRA

$$\frac{1}{1 - 0.3709} = \frac{1}{0.6291} = 1.5896 \text{ (accumulation factor at } \tau = 6 \text{ hr)}$$

$e^{-k\tau}$ = 0.3709 from part g. With a dose every 6 hours, 37% of each dose is still present when the next arrives, so the peak builds to 1.59 times the single-dose value.

ALGEBRA

$$e^{-0.1653 \times 2.013} = e^{-0.3327} = 0.7170$$

t_{max}^{∞} = 2.013 hr from part g. $e^{-k\, t_{max}^{\infty}}$ is the fraction left after elimination has run from the dose to the time of the peak, so the oral peak sits below 80 × 1.5896.

ALGEBRA

$$C_{max}^{\infty} = 80 \times 1.5896 \times 0.7170 = 91.18 \text{ mg/L}$$

Product of the three factors: 80 mg/L × 1.5896 = 127.17 mg/L, then × 0.7170 = 91.18 mg/L. The units come from the first factor; the other two are pure numbers.

ROUND

$$C_{max}^{\infty} = 91.2 \text{ mg/L}$$

With τ = 6.05 hr unrounded the peak is 90.6 mg/L; her key may use either interval, and the tolerance here accepts both. Report it with units, mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> A steady-state peak lies above the 80 mg/L average the regimen was designed for: 91.2 mg/L.

Question 4

Worked by the drill

Answer: 8.9 mg/L

WHAT THE STEM GIVES

D_0	Four hundred milligrams	$R = \frac{400}{2} = 200$ mg/hr
tinf	2 hours	infusion time in $1 - e^{-kt}$
second dose	Eight hours after the start	infusion 2 runs 8 to 10 hr
t½	approximately 3.75 hours	k = 0.1848 hr^{-1}
V_D	approximately 22 L	$Cl_T = kV_D$ = 4.066 L/hr
t	4 hours after the cessation	the level at 14 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Infusion 1 runs 0 to 2 hr; infusion 2 runs 8 to 10 hr; the level is asked at 10 + 4 = 14 hr

Draw the timeline first. At 14 hr, infusion 1 has been declining for 14 − 2 = 12 hr and infusion 2 for 14 − 10 = 4 hr.

UNIT

$k = \frac{0.693}{3.75\ \text{hr}} = 0.1848\ \text{hr}^{-1}$; $Cl_T = k \times V_D = 0.1848 \times 22\ \text{L} = 4.066\ \text{L/hr}$; $R = \frac{400\ \text{mg}}{2\ \text{hr}} = 200\ \text{mg/hr}$

R is the infusion rate, the dose divided by the hours it runs: 400 mg over 2 hr. k comes from the half-life and Cl_T from $k \times V_D$, because the stem gives neither directly.

ALGEBRA

$C_{end} = \frac{R}{Cl}(1 - e^{-kt}) = \frac{200}{4.066}(1 - e^{-0.1848 \times 2}) = 49.19 \times (1 - 0.6910) = 49.19 \times 0.3090 = 15.20\ \text{mg/L}$

The level at the end of one 2-hour infusion. Both infusions are the same dose, rate and length, so each reaches 15.20 mg/L when it stops.

ALGEBRA

Infusion 1: $15.20 \times e^{-0.1848 \times 12} = 15.20 \times 0.1089 = 1.655\ \text{mg/L}$

After an infusion stops, only first-order elimination: $C = C_{end}\, e^{-kt}$, t counted from the end of that infusion.

ALGEBRA

Infusion 2: $15.20 \times e^{-0.1848 \times 4} = 15.20 \times 0.4775 = 7.258\ \text{mg/L}$

4 hours after the second infusion stopped, at 14 hr. Four hours is just over one half-life of 3.75 hours, so a little under half of the 15.20 mg/L is left.

ALGEBRA

$C_p = 1.655 + 7.258 = 8.913\ \text{mg/L}$

Superposition: concentrations from separate infusions add, because elimination is first order and each dose is removed at the same fractional rate whatever else is present.

ROUND

$C_p = 8.9\ \text{mg/L}$

Her Quiz 4, question 4, was this stem with t½ 4 hours and V_D 20 L, key 10.6 mg/L. Report the answer with units, mg/L, to the decimal places the item asks.

> **DOES THE NUMBER MAKE SENSE?**
> Infusion 2 alone leaves under its 15.20 mg/L end level after 4 hours, about half of it since 4 hours is near one half-life; infusion 1 adds a little: 8.9 mg/L.

Question 5

Worked by the drill

Answer: A. renal function

Creatinine is filtered by the glomerulus, so its clearance estimates the glomerular filtration rate, the usual measure of how well the kidneys work. Her slide gives 120 to 130 mL/min as normal.

Question 6 Worked by the drill

Answer: B. Increased AUC; F. Increased C_{max}

AUC = $\frac{F\ D_0}{Cl}$: the dose is in the numerator and nothing else changes, so the area doubles. Her slide "Changing Dose" shows AUC rising in proportion to the dose.

D_0 multiplies the whole curve, $C_p = \frac{F\ k_a\ D_0}{V_D(k_a - k)}(e^{-kt} - e^{-k_a\ t})$, so the peak doubles along with every other point while its time stays the same.

Question 7 Worked by the drill

Answer: B. Filtration with active secretion

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R \approx 300$ mL/min; GFR ≈ 120 mL/min

Renal clearance is compared with the glomerular filtration rate, the most plasma filtration alone can clear each minute.

ALGEBRA

$\frac{300\ \text{mL/min}}{120\ \text{mL/min}} = 2.5$, so Cl_R is greater than GFR: filtration with active secretion

Greater than GFR needs a second process adding drug to the urine, because filtration alone clears at most the volume the glomeruli filter each minute, about 120 mL/min.

300 mL/min is above the glomerular filtration rate of about 120 mL/min. Filtration cannot clear more plasma than it filters, so a transporter must be carrying extra drug from the blood into the tubule.

Question 8 Worked by the drill

Answer: B. bioavailability

This is her slide definition of bioavailability word for word: rate and extent of absorption from a drug product to the site of action.

Question 9 Worked by the drill

Answer: A. increased steady-state concentrations

Her slide "Altering Dosing Interval": decreasing the dosing interval increases steady-state concentrations. Each dose arrives before as much of the last has left, so C_{max}^{∞}, C_{min}^{∞} and C_{avg}^{∞} all rise.

Question 10 Worked by the drill

Answer: A. True

Her slide "Superposition" lists two assumptions: the drug is eliminated by first-order kinetics, and the pharmacokinetics of the drug after a single dose are not altered after multiple doses. The statement is the second one.

Question 11 Worked by the drill

Answer: D. absolute bioavailability

Her slide "Absolute Bioavailability" word for word: extravascular compared with intravenous. The IV dose is the reference because all of it reaches the systemic circulation.

Question 12 Worked by the drill

Answer: B. the rate of drug absorption is faster than the rate of drug elimination

Before the peak more drug enters from the absorption site each hour than is eliminated, so the concentration rises. That rising limb is the absorption phase.

Question 13 Worked by the drill

Answer: A. at the end of the dosing interval

A bolus puts the whole dose in at once, so the level is highest just after each dose and falls until the next one. The lowest point is just before the next dose, the end of the interval.

Question 14 Worked by the drill

Answer: A. the elimination half-life will increase

$t½ = \frac{0.693 \times V_D}{Cl}$. With V_D unchanged, lower clearance means slower removal, so it takes longer for half of the drug to leave.
